



UNIVERSITY OF SCIENCE  
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# Writing Paper's Method Section

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# Outline

- Proposed approach
- Experimental design
- Exercise

# Typical Paper Organization

- Title, along with names, affiliations, email addresses of authors
- Abstract
- Introduction
- Background
- Related work
- **Proposed approach**
- Evaluation
  - Experimental design
  - Results
  - Discussions
- Threats to validity (optional)
- Conclusions

# Paper body's organization – two common options (1/2)

- Option 1: separate proposed approach section
  - **Proposed approach**: details your approach
  - **Experimental design**: details the method to evaluate the proposed approach
  - **Results**: presents results
  - **Discussions**: provides discussions on the results
  - **Threats to validity**: discusses possible threats to validity (if any)

This option is suitable for papers proposing a new approach, technique, framework, or tool

# Paper body's organization – two common options (2/2)

- Option 2: NO separate proposed approach section
  - **Methodology**: details the study to address the problem
  - **Results**: presents results
  - **Discussions**: provides discussions on the results
  - **Threats to validity**: discusses possible threats to validity (if any)

This option is suitable for papers presenting controlled experiments to evaluate existing approaches, case studies, empirical studies, replicated studies, systematic reviews, etc.

# Proposed approach (1/4)

- This section presents your **new** approach, technique, framework, or tool
- Many papers do not propose any new model, method, algorithm, etc.
  - Most case studies, replicated studies, empirical studies do not introduce new approaches
- But they need to have an approach for the present study
  - Approach for case studies, replicated studies
- Heading names: “The Approach”, “Proposed Approach”, “Our Approach”, or a specific naming to present your approach
- May include illustrating or motivating examples to demonstrate your approach

# Proposed approach (2/4)

- Illustrating/motivating examples

- Examples

- Diagrams, figures, code listings
    - Should be simple
    - Should be typical to the problem you are addressing

- Should present how your approach can be applied to the examples

- Examples can be included in the “Proposed approach” section or a separate section prior to this section

- Throughout the paper, you can relate your approach to the example(s)

# Proposed approach (3/4)

- Generalize your work in an abstraction level, e.g., positioning it as an approach/framework rather than a tool
  - What you develop should be beyond your own implementation
  - Then you are in a better position when you discuss limitations of your work
    - Inherent limitation of the framework?
    - Or limitation of your current particular implementation of the framework?
  - A workflow diagram is useful for explaining your framework
- Try to separate the ideas from (a particular) concrete implementation
  - But sometimes you have to mention it a bit and refer the readers to the implementation section

*Slide adapted from Tao Xie*



# Proposed approach (4/4)

- What libraries you use in your tool/framework
- Detailed implementations of each step in your framework
- List complications of implementing a certain idea and how you get around them
  - If some complications are important and general, you may move them to the framework section
- Generally, provide sufficient details in your approach so that readers can distinguish between your work and previous work

*Slide adapted from Tao Xie*

# Outline

- Proposed approach
- Experimental design
- Exercise

# Experimental design (1/6)

- Provides details of the method designed to evaluate the proposed approach
  - How and why the experiment is conducted
  - The relationship between the experiment and other sections
  - How to replicate the study
  - How to evaluate the results
- It is very important to provide all details needed to make sure others can **replicate** the experiment
- Typical headings
  - “Evaluation”, “Methods”, “Experimental design”, “Experimental setup”
  - ...

# Experimental design (2/6)

- This section provides the following
  - Research questions/hypotheses
  - Procedure used to address the research questions/hypotheses
  - Subjects, materials
  - Environments: software (e.g., IDEs, tools, libraries used) and hardware (computers, devices)
  - Assessment measures (e.g., performance metrics)
  - Data collection
  - Data analysis (including statistical method, software package used, e.g., Python)
    - Independent variables and dependent variables

# Experimental design (3/6)

- Typical structure
  - Research questions/hypotheses
  - Subjects/participants/groups
  - Procedure
    - Used to address the research questions/hypotheses
    - Data collection/data sets
    - Data analysis (including statistical method, software package used, e.g., Python)
      - independent variables, dependent variables
  - Environment setup
    - Software (e.g., IDEs, tools, libraries used) and hardware (computers, devices)
  - Metrics
    - Assessment measures

# Experimental design (4/6)

- You can have research questions or hypotheses or both
- If both, research questions have to be written before hypotheses
- Research questions: abstract, general, high-level
- Hypotheses: concrete, specific
- Each research question/hypothesis has to be explained
- Every research question/hypotheses have to be discussed in the results section

# Examples of research question

## 5.1 Research questions

*5.1.1 RQ1. How does the proposed method perform in comparison to existing approaches to UI test case prioritization?* We include several existing methods in the literature. These methods are related to the proposed approach in terms of concepts and techniques used which will be further discussed in Section 5.3.

*5.1.2 RQ2. How much test is needed for achieving 100% test coverage?* We evaluate the practicality of the approach by investigating the reduction in the time needed for test suite execution. In other words, we find the percentage of test suite execution needed for 100% faults coverage when applying the method and compare the results with related approaches.

An Nguyen, Bach Le, Vu Nguyen.  
“Prioritizing automated user interface tests using reinforcement learning”.  
The 15th International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE 2019). September, 2019. Porto de Galinhas, Brazil

# Examples of hypothesis

## 4.1. Hypotheses

According to Boehm [7], programmer's maintenance activities consist of understanding maintenance task requirements, code comprehension, code modification, and unit testing. Although the last two activities deal with source code directly, empirical studies have shown high correlations between the overall maintenance effort and the total LOC added, modified, and deleted (e.g., [2,21]). We hypothesize that these activities have comparable distributions of programmer's effort regardless of what types of changes are made. Indeed, with the same cost factors [5] such as program complexity, project and personnel attributes, the productivity of enhance tasks is expected to have no difference with that of corrective and reductive maintenance. Thus, we have the following hypotheses:

**Hypothesis 1.** There is no difference in the productivity among enhance, corrective, and reductive maintenance.

**Hypothesis 2.** There is no difference in the division of effort across maintenance activities.

Vu Nguyen, Barry W. Boehm,  
Phongphan Danphitsanuphan, "A  
Controlled Experiment in Assessing and  
Estimating Software Maintenance  
Tasks." Journal of Information and  
Software Technology, June 2011



# Experimental design (5/6)

## ■ Subjects/participants/groups

- ❑ Necessary of your study involves human subjects (participants in your experiment)
- ❑ How do you select participants?
- ❑ Do you divide into groups? How?
- ❑ Participant selection affects the generalizability (*external validity*) of the study
  - ❑ e.g., you develop a tool for developers to use, to evaluate your tool you experiment it using all first-year students. Is this a concern?
  - ❑ e.g., you develop a tool for doctors to use, you evaluate it using students. Is this a concern?

# Experimental design (6/6)

## ■ Procedure

- ❑ Detail all steps you have taken, should be in actual sequence of events
- ❑ Write each group of events in one or several paragraph
- ❑ Describe methods for data collection
- ❑ Describe data sets (you may separate data sets into subsections)
- ❑ Make sure your procedure can be reproduced

## ❑ Data analysis

- Describe statistical methods used, hypothesis testing
  - ❑ E.g., t-test, Anova test
- Statistical package/tool (e.g., R, Python)
- Independent variables and dependent variables

# Examples of hypotheses

- Comparing two different algorithms for sorting, quick and bubble sorts
  - Hypotheses
    - H1: There is no difference in the time taken to run between the quick and bubble sort algorithms.
    - H2: There is no difference in the memory used by the quick and bubble sort algorithms.
  - Dependent variables
    - H1? Time
    - H2? Memory usage
  - Independent variables
    - H1: type of sort (quick sort, bubble sort)
      - size of array, computers, etc. → isolated.
    - H2: type of sort (quick sort, bubble sort)
      - size of array, computers, etc. → isolated.

# Exercise 1

- Work with your partner to write section headings for your paper
  - Propose an outline for your paper
- The outline has to be suitable for your topic and type of paper
  
- Draft the introduction section of your paper. Make sure it contains the following information
  - Context, Problem and its importance
  - Related studies addressing the problem
  - Proposed solutions/ideas to address the problem
  - Results, contributions
- Time: 30 minutes

## Exercise 2

- Draft the methodology or proposed approach of your paper
  - Draw diagrams to present the idea
- Define research questions for your study
- Present them to the class
- Time: 60 minutes